

Note – a Special Case for $N^A = 3$

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According to the Yao-Qi method, the entanglement entropy of a subregion A of the Kitaev model can be expressed as

$$S_A = -\text{tr} \left(\frac{\Gamma_A}{2} \ln \frac{\Gamma_A}{2} \right) + \left(\frac{1}{2} |\partial A| - 1 \right) \ln 2 \quad (1)$$

where Γ_A is a $N^A \times N^A$ matrix defined by

$$[\Gamma_A]_{ij} = \langle c_i c_j \rangle \quad ; \quad i, j \in A$$

If we take $N^A = 3$, and define $\langle c_1 c_2 \rangle = ix$, $\langle c_3 c_2 \rangle = iy$, and $\eta = \sqrt{x^2 + y^2}$, then the matrix Γ_A reads

$$\Gamma_A = \begin{pmatrix} 1 & ix & 0 \\ -ix & 1 & -iy \\ 0 & iy & 1 \end{pmatrix}$$

The three eigenvalues of such a matrix are $\{1, 1 \pm \eta\}$. We also know that the length of the circumference of A is $|\partial A| = 5$. Thus, equation (1) gives the entanglement entropy as following

$$S_A = 3 \ln 2 - \frac{1}{2} [(1 + \eta) \ln (1 + \eta) + (1 - \eta) \ln (1 - \eta)] \quad (2)$$

On the other hand, thanks to the small size of A we chose, the entanglement entropy can be calculated via exactly diagonalizing the reduced density matrix. The Majorana fermion representation of the reduced density matrix is

$$\rho_A^F = \frac{1}{4} (1 + \langle ic_1 c_2 \rangle ic_1 c_2 + \langle ic_3 c_2 \rangle ic_3 c_2) = \frac{1}{4} (1 - xic_1 c_2 - yic_3 c_2)$$

The factor $\frac{1}{4}$ corresponds to $2^{-\lceil N^A/2 \rceil}$. Our next goal is to rearrange the formula into the Gibbs form $\mathcal{K}e^{-H_{\text{ent}}}$ to get the entanglement Hamiltonian H_{ent} . In this way, we can obtain the full information of the entanglement spectrum, and naturally get the value of the entanglement entropy. To do this, we define $\gamma_1 = \frac{x}{\eta}c_1 + \frac{y}{\eta}c_3$ and $\gamma_2 = c_2$. After that, the Majorana reduced density matrix becomes

$$\rho_A^F = \frac{1}{4} [1 + \eta (-i\gamma_1 \gamma_2)] = \frac{1}{4 \cosh \epsilon} [\cosh \epsilon + \sinh \epsilon (-i\gamma_1 \gamma_2)] = \frac{1}{4 \cosh \epsilon} \exp(-i\epsilon \gamma_1 \gamma_2)$$

where we have defined ϵ in the way such that $\tanh \epsilon = \eta$. The corresponding entanglement entropy is then

$$S_A^F = -\text{Tr}_F (\rho_A^F \ln \rho_A^F) = \epsilon \langle i\gamma_1 \gamma_2 \rangle + \ln 4 + \ln \cosh \epsilon \quad (3)$$

Represent γ_1 and γ_2 in terms of c -Majorana fermions, we get

$$\langle i\gamma_1 \gamma_2 \rangle = \frac{x \langle ic_1 c_2 \rangle + y \langle ic_3 c_2 \rangle}{\eta} = -\frac{x^2 + y^2}{\sqrt{x^2 + y^2}} = -\eta$$

On the other hand, the term $\ln \cosh \epsilon$ can also be simplified,

$$\ln \cosh \epsilon = -\frac{1}{2} \ln (1 - \tanh^2 \epsilon) = -\frac{1}{2} [\ln (1 + \eta) + \ln (1 - \eta)]$$

At the same time, ϵ can also be represented in terms of η in a simple way,

$$\epsilon = \tanh^{-1} \eta = \frac{1}{2} [\ln(1 + \eta) - \ln(1 - \eta)]$$

Put all of these expressions back to (3), we obtain

$$S_A^F = 2 \ln 2 - \frac{1}{2} [(1 + \eta) \ln(1 + \eta) + (1 - \eta) \ln(1 - \eta)] \quad (4)$$

What we have obtained is merely the fermion part of the entanglement entropy. To obtain the correct value of the entanglement entropy, we have to add the corresponding gauge contribution back, which is

$$S_A^G = \left(N^A - \left\lceil \frac{N^A}{2} \right\rceil - N_p^A \right) \ln 2 = \ln 2$$

So, eventually, we get

$$S_A = S_A^F + S_A^G = 3 \ln 2 - \frac{1}{2} [(1 + \eta) \ln(1 + \eta) + (1 - \eta) \ln(1 - \eta)] \quad (5)$$

which reproduces the value obtained by the Yao-Qi formula (2).

The gauge contribution S_A^G arises from the degeneracy of the entanglement spectrum

$$\rho_A = \frac{1}{2^{N^A}} (\mathbb{1} + \langle \sigma_1^x \sigma_2^x \rangle \sigma_1^x \sigma_2^x + \langle \sigma_3^y \sigma_2^y \rangle \sigma_3^y \sigma_2^y) \cong \frac{1}{2^{N^A - \lceil N^A/2 \rceil}} \begin{pmatrix} \rho_A^F & \\ & \rho_A^F \end{pmatrix}$$

We can see the factor $\frac{1}{2^{N^A - \lceil N^A/2 \rceil}}$ counts the degeneracy, and which contributes to S_A^G we have obtained.

To confirm this, we can again diagonalize the density matrix ρ_A . Such a density matrix can be written as

$$\rho_A = \frac{1}{8} \left[\mathbb{1} - \eta \left(\frac{x}{\eta} \sigma_1^x \sigma_2^x + \frac{y}{\eta} \sigma_3^y \sigma_2^y \right) \right] = \frac{1}{8 \cosh \epsilon} \left[\cosh \epsilon \mathbb{1} - \sinh \epsilon \left(\frac{x}{\eta} \sigma_1^x \sigma_2^x + \frac{y}{\eta} \sigma_3^y \sigma_2^y \right) \right]$$

and one can observe that the operator $\left(\frac{x}{\eta} \sigma_1^x \sigma_2^x + \frac{y}{\eta} \sigma_3^y \sigma_2^y \right)$ is an $\mathbb{Z}_2$ operator.

$$\left(\frac{x}{\eta} \sigma_1^x \sigma_2^x + \frac{y}{\eta} \sigma_3^y \sigma_2^y \right)^2 = \frac{1}{\eta^2} (x^2 \mathbb{1} + y^2 \mathbb{1}) = \mathbb{1}$$

So the density matrix can be written as the Gibbs form

$$\rho_A = \frac{1}{8 \cosh \epsilon} \exp \left[-\epsilon \left(\frac{x}{\eta} \sigma_1^x \sigma_2^x + \frac{y}{\eta} \sigma_3^y \sigma_2^y \right) \right]$$

One should notice that there's no N_p^A in the formula because A is too small so that there's no plaquette exclusively inside A , thus $N_p^A = 0$ in the case of $N^A = 3$. If we consider larger subregions, we have to include the projector $\mathcal{P}_A = \prod_{p \in A} \left(\frac{\mathbb{1} + W_p}{2} \right)$ inside the reduced density matrix ρ_A , and the factor N_p^A will appear naturally.